

Chapter 8

Important Libraries for AI

In this chapter we will cover all the basics libraries and some frameworks of machine learning, deep learning and computer vision. We will discuss all about their use with an example and instruction which will states the commands to install to those libraries for building any project in our system. So let's begin

What is a Library?

Library is an aggregate of features and methods that allow you to take many actions without getting your own code written.

For instance, the libraries that you have to know are if you work with data, numpy, scipy, pandas, etc. They have very simple data processing functions that save you time for small tricks.

Machine Learning and Python

- **TensorFlow:** Google, along with the team of Brain, created this library. For almost every Google machine learning program, TensorFlow is used. As neural networks can be easily represented as computer graphs, they can be used as a Tensor operation using the TensorFlow method, TensorFlow operates like a code-library for writing new algorithms involving a number of tensor operations. Moreover, tensors represent the data in N-dimensional matrices.
- **Features of TensorFlow:** For fast linear algebra operations, Tensor Flow is optimized, employs techniques like XLA.
 - **Responsive Construct:** With TensorFlow, each part of the graph that is not an option when using NumPy or SciKit can be visualized easily.
 - **Flexible:** One of the most important aspects of TensorFlow is that its operability is versatile, which means it has modularity and the parts of it you want to render independently.
 - **Easy Trainable:** On CPU and GPU for distributed computing it is easily trained.
 - **Parallel Neural Network Training:** TensorFlow provides pipelining to train multiple neural networks and multiple GPUs, making modeling on large-scale systems extremely effective.
 - **Large Community:** Naturally, a large team of software engineers has formed by Google that works continuously on stabilization enhancements.
 - **Opensource:** The great thing about this learning machine library is that it is open source so that anyone can use it when they want just by downloading it from online.

- Command to Install Scikit-learn

Pip install tensorflow

Example:

```
# Python program using TensorFlow
# for multiplying two arrays

# import `tensorflow`
import tensorflow as tf

# Initialize two constants
x1 = tf.constant ([1, 2, 3, 4])
x2 = tf.constant ([5, 6, 7, 8])

# Multiply
result = tf.multiply (x1, x2)

# Initialize the Session
sess = tf.Session ()

# Print the result
print (sess.run (result))

# Close the session
sess.close ()
```

Output:

```
[ 5 12 21 32]
```

- **Scikit-learn:** It is a Python library with NumPy and SciPy connections. It is recognized as one of the best libraries for complex data processing. It is built on Numpy, Scipy and Matplotlib.

This library includes a number of improvements. One improvement is the cross-validation function that allows more than one metric to be used. Some training techniques such as regression of logistics and closest neighbors have been enhanced.

- Features of Scikit-Learn

Cross Validation: Specific techniques for the accuracy of tracked models on unsightly data are available

Unsupervised Learning Algorithm: A broad range of algorithms – from clustering, factor analysis, main component analysis to unsupervised neural networks – are available in the package.

Feature Extraction: Useful for extracting image and text features (e.g. word bag)

- Command to Install Scikit-learn

Pip install scikit-learn

- **NumPy:** It is considered to be one of Python's most popular machine Learning library. For multiple operations on tensors, TensorFlow and other libraries internally use NumPy. NumPy's best and main feature is the Array interface.

- Features of NumPy:

Interactive: NumPy is very interactive and user-friendly.

Mathematics: Makes complex math very simplistic.

Intuitive: It is quick to code and it is easy to understand the concepts.

Lot of Interaction: A lot of open source participation, therefore, it is widely used.

Command to Install NumPy –

Pip install numpy

Example for NumPy: (Save the File as Num1.py)

```
# Python program using NumPy
# for some basic mathematical
# operations

import numpy as np

# Creating two arrays of rank 2
x = np.array ([[4, 5], [6, 8]])
y = np.array ([[8, 9], [10, 11]])

# Creating two arrays of rank 1
v = np.array ([12, 13])
w = np.array ([15, 16])

# Inner product of vectors
print (np.dot (v, w), "\n")

# Matrix and Vector product
print (np.dot (x, v), "\n")

# Matrix and matrix product
print (np.dot (x, y))
```

Output:

```
>>>
```

```
388
```

```
[113 176]
```

```
[[ 82 91]
 [128 142]]
>>>
```

- **SciPy:** It is an application creation and engineering machine learning library. Nonetheless, the gap between SciPy and SciPy stack still has to be identified. SciPy library contains optimization modules, linear algebra, integration modules and statistics.

Features of SciPy:

The key advantage of the SciPy library is that NumPy is used for the creation of the array.

In addition, with its unique substructures SciPy offers all the powerful numerical routines such as optimization, numerical integration and many others.

Command to Install scipy –

Pip install scipy

- **Pandas:** Pandas is a Machine Learning library in Python which offers high-level data structures and a wide range of analytical tools. The ability to translate complex operations with data using one or two commands is one of the major features of this library. Pandas have many standardized grouping methods, data combination and filtering methods, as well as functions for time series. Those are all accompanied by excellent speed measurements.
- Features of pandas: Pandas will make it easier to manipulate data throughout. Pandas' function highlights provide support for operations such as re-indexing, Iteration, Sorting, aggregations, concatenations and visualizations.

Command to Install pandas –

Pip install pandas

- **Matplotlib:** Matplotlib is a popular data visualization library for Python. It's not related to machine learning directly, like Pandas. It is especially useful to imagine patterns in the data if a programmer wants to. This is a library for 2D plotting used for the development of 2D plots. A module called pyplot makes it easy for plotting programmers to control line styles, fonts, axes, etc. It offers different types of graphs and plots for viewing data, e.g. histograms, error charts and chats, etc.

Command to Install Matplotlib –

Pip install matplotlib

Example

```
# Python program using Matplotlib
# for forming a linear plot

# importing the necessary packages and modules
import matplotlib.pyplot as plt
import numpy as np
```

```
# Prepare the data
x = np.linspace (0, 10, 100)

# Plot the data
plt.plot (x, x, label ='linear')

# Add a legend
plt.legend ()

# Show the plot
plt.show ()
```

Output: as Fig. 8.1

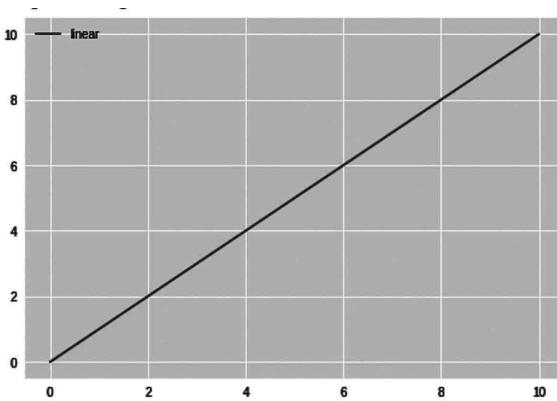


Fig. 8.1 Program output

- **Keras:** It is considered as one of the best machine learning libraries in Python. This offers an simpler way to describe neural networks. Keras provides some of the best tools for model creation, data-sets analysis, graph visualization and much more. Keras uses Theano or TensorFlow internally in the background. This may also use some of the best-known neural networks as CNTK. When comparing it with other learning machine libraries, Keras is comparatively sluggish. Since it uses back-end technology to generate a data graph and then uses it to perform operations. In Keras, all models are mobile.

Feature of Keras:

- On CPU and GPU operate smoothly.
- Keras supports virtually all neural network models-completely connected, convolutionary, pooling, recurrent, convergence, etc. Such models can also be combined to create more complex models.
- The modular design of Keras is incredibly articulate, versatile and perfect for creative work.

- Keras is a Python-based framework that makes debugging and exploring simple. Command to Install Keras –

Pip install Keras

- **PyTorch:** It is the biggest machine educational library which enables developers to perform tensor calculations by accelerating the GPU, creating dynamic graphs and automatically calculating gradients. However, PyTorch provides a wide range of APIs for the resolution of neural network device issues. This machine Learning library is developed in the form of Torch, an open source software library in C with a Lua wrapper. Released in 2017, this library in Python became popular and attracts a growing number of machine learning developers since its inception.

Features of PyTorch:

- Hybrid Front End: A new hybrid front end provides ease of use and versatility in the eager mode while transitional in C++ operating environments into a graphic mode for speed, optimization and functionality.
- Distributed Training: Optimize research and development efficiency by benefiting from native assistance to a synchronize collective operations and peer-to-peer networking accessible from Python and C++

Command to Install Pytorch –

Pip install torchvision

- **Theano:** It is a Python computational framework library for multidimensional array computing. TensorFlow functions similarly to Theano, though not as strong as TensorFlow. Due to its inability to adapt to manufacturing environments. Theano can be used in TensorFlow-like distributed or parallel environments

Features of Theano:

- Tight integration with NumPy: Capacity in Theano-compiled functions to use completely NumPy arrays.
- Transparent use of a GPU: Perform computations with data intensity much more quickly than a CPU.
- Efficient symbolic differentiation: For functions of one or more inputs, Theano does the derivatives.
- Speed and stability optimizations: Get the right log (1+x) answer even if x is very small. It is only one example of Theano's stability.
- Dynamic C code generation: Evaluate terms faster than ever, thus making them much more effective.
- Extensive unit-testing and self-verification: Detect and diagnose many types of models of errors and ambiguities.

Command to Install Theano –

Pip install Theano

Deep Learning

- **Mxnet:** It is a deep learning open-source framework used for neural networking

and training. It is highly productive and fast. It can scale through several GPUs and several machines.

To Install Maxnet – Download the package and install the package

https://mxnet.apache.org/get_started/

Link for more details: <https://mxnet.apache.org/>

- **Caffe:** Caffe is a deep learning framework made with expression, speed, and modularity in mind. It is developed by Berkeley AI Research (BAIR) and by community contributors.

To Install Caffe– prefer the documentation on this link

<https://caffe.berkeleyvision.org/installation.html>

Link for more details: <https://caffe.berkeleyvision.org/>

- **Fast.ai:** The fastai library simplifies training fast and accurate neural nets using modern best practices. It's based on research in to deep learning best practices undertaken at fast.ai, including “out of the box” support for vision, text, tabular, and collab (collaborative filtering) models.

To Install Fast.ai– prefer the documentation on this link

<https://github.com/fastai/fastai>

Link for more details:<https://docs.fast.ai/>

- **CNTK:** A toolkit for commercial-grade distributed deep learning is the Microsoft Cognitive Toolkit (CNTK). Neural networks are defined via a guided chart as a series of computer measures. CNTK makes it simple to incorporate and combine common models such as feed-forward DNNs, CNNs and recurring neural networks (RNNs/LSTMs). CNTK allows automatic differentiation, and parallelization, of stochastic gradient descent (SGD) learning through multiple GPUs and servers.

CNTK supports 64-bit Linux or 64-bit Windows operating systems. To install you can either choose pre-compiled binary packages, or compile the toolkit from the source provided in GitHub:<https://github.com/Microsoft/CNTK>

Link for more details:<https://docs.microsoft.com/en-us/cognitive-toolkit/>

ComputerVision

- **OpenCV:** OpenCV is one of the most widely used libraries for computer vision applications. Python API for OpenCV is the OpenCV Python. OpenCV-Python is not only easy, as the context of the C/C++ code is easily coded and deployed (as a result of Python wrapper). This makes the execution of computer-intensive vision programs a good choice.

Command to Install OpenCV–

Pip install opencv-python

- **Imutils:** The simple image processing tasks such as translation, rotate, resize, skeletonization, display, Matplotlib images, contours sorting and edges are simplified with OpenCV and Python in a variety of convenience features.

Command to Install imutils –

Pip install imutils

- **Pillow/PIL (python Imaging Library):** PIL (Python Imaging Library) is a free Python programming language library that supports several different image file formats for opening, handling, and saving. Nevertheless, with its last release in 2009, its growth has stagnated. Thankfully, Pillow, an popular PIL fork, is more easily installed, running on all major operating systems, and supporting Python 3. The library has simple image processing features, such as point operations, filtering with built-in cooling kernels and conversions of color space.

Command to Install pillow –

Pip install pillow

- **Scikit-image:** Scikit-image is the Python programming language open-source image processing library. The package includes segmentation algorithms, geometrical transformations, color space manipulation, visualization, filtering, morphology, function detection and much more. Scikit-image is a series of image processing algorithms. This is open and free of any restrictions. This requires some implementations of algorithms that OpenCV does not.

Command to Install Scikit-image –

Pip install Scikit-image

Link for more details:<https://scikit-image.org/>

- **SimpleCV:** Another open source platform for computer vision applications is SimpleCV. It allows access to various powerful computer vision libraries like OpenCV without knowing bit depth, file formats, color spaces, and so on. The curve of his education is slightly smaller than that of OpenCV and (as his slogan says) “it’s easy to see a machine.”

Link for more details:<http://simplecv.org/>

- **Vigra:** It is a computer vision library that focuses on scalable algorithms as the key know-how of algorithms in this area. As a result, the book was build in the standard template library, using generic programming pioneered by Stepanov and Musser. You can use VIGRA algorithms in addition to your data structures within your environment by writing a few adapters (image iterators and accessories). Also, you can use VIGRA data structures that are easy to adapt to a wide variety of applications. VIGRA has practically no versatility, as the architecture uses polymorphism (templates) for compile time

Link for more details: <https://ukoethe.github.io/vigra/>

- **Mahotas:** It is another library for Python’s computer vision and picture processing. It includes conventional image treatment functions, such as filters and morphological operations, as well as more modern machine viewing functions, including interest point detection and local descriptors for object computation. The Python code is ideal for rapid development, but the algorithms are used in C++ and speed tuned. The library of Mahotas is fast with minimal code and even minimum dependency.

Link for more details:<https://mahotas.readthedocs.io/en/latest/>

- **Pycairo:** Pycairo is a Python module that supplies the library with bindings. It works with Python 3.5 + and relies on cairo > = 1.13.1. Pycairo is authorised under the LGPL-2.1-only OR MPL-1.1 including this report. The Pycairo linking systems have been designed to suit the Cairo API as closely as possible, and only deviate if explicitly more 'pythonically' deployed.

Featrures of pycairo:

- Offers a cairo-focused object gui.
- Searches and converts error status of artifacts to exceptions.
- The C API is provided so other Python extensions can be used.

Link for more details:<https://pypi.org/project/pycairo/>

Pgmagick: For the Graphics Magick library, the pgmagick is a Python-based wrapper. The image treatment system Graphics Magick is sometimes referred to as the Swiss Army Knife. The powerful and effective range of images in over 88 major formats, including DPX, GIF, JPEG, JPEG-2000, PNG, PDF, PNM and TIFF, facilitates the reading, writing and editing of photos.

Link for more details:<https://pypi.org/project/pgmagick/>

Summary

Here we have seen some of the important libraries of these domains of artificial intelligence and also have covered example for some of them with justify their use. There are so many libraries for python which can be useful for different projects to built in artificial intelligence and other area of study for different perspective and objective.